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Experiment No. 5: Demonstration of switch hardware and setting up of basic LAN

Aim

To identify the hardware interfaces of a Switch and to design, configure, and verify a basic Local Area Network (LAN) using a Star Topology to observe broadcast behavior in Cisco Packet Tracer.

Theory

1. **What is a Switch?**
   A switch is an intelligent networking device used to connect multiple computers or devices in a network. It operates at the Data Link Layer (Layer 2) of the OSI model. A switch filters and forwards data based on MAC addresses, ensuring that data is sent only to the intended device instead of all devices.

2. **Working Principle of a Switch.**
   A switch works on the principle of selective forwarding (unicast communication). When a device sends data:

1. The switch receives the data frame.
2. It checks the destination MAC address.
3. It refers to its MAC address table (CAM table).
4. It forwards the data only to the specific port where the destination device is connected.
5. If the MAC address is not in the table, it broadcasts the frame initially and then learns the address for future communication.
   This reduces network congestion and minimizes data collisions.

3. MAC Address

A MAC (Media Access Control) address is a unique physical address assigned to a network interface card (NIC) of a device.
- It is a 12-digit hexadecimal number.
- Example: 00:1A:2B:3C:4D:5E
- It works at the Data Link Layer (Layer 2) of the OSI model.
- It is permanently assigned by the manufacturer.

It is used by switches to identify devices and build MAC address tables.

4. IP Address

An IP (Internet Protocol) address is a logical address assigned to a device in a network to identify it and enable communication.
- It works at the Network Layer (Layer 3).
- It can be changed.

• Example of IPv4: 192.168.1.1
• Example of IPv6: 2001:0db8:85a3::8a2e:0370:7334
It helps in locating devices across different networks.

5. Local Area Network (LAN)

A Local Area Network (LAN) is a network that connects computers and devices within a small geographical area such as a home, school, office, or college.
• It allows sharing of resources like files, printers, and internet.
• It provides high-speed communication.
• It is privately managed.

6. Star Topology

A star topology is a network layout in which all devices are connected to a central device such as a hub or switch.
• Each device has a separate connection to the central device.
• If one cable fails, only that device is affected.
• If the central device fails, the entire network stops working.
It is easy to install and troubleshoot and is commonly used in LANs.

Hardware Components Required

• PC-PT (x3): End devices acting as network nodes.
• Switch-PT: A Data Link Layer device used to interconnect the nodes.
• Copper Straight-Through Cables: The transmission media used for PC-to-Switch connectivity.

Hardware and Connection Setup
• Network ID: 192.168.1.0
• Subnet Mask: 255.255.255.0
• PC Naming Convention:
o PC 1: Batch_Roll_A (e.g., S11_01_A) o
PC 2: Batch_Roll_B (e.g., S11_01_B) o
PC 3: Batch_Roll_C (e.g., S11_01_C)

Procedure

Step 1: Physical Topology Design

1. Open Cisco Packet Tracer and select Network Devices > Switches > Switch-PT. Place it in the center.

2. Select End Devices > PC and place four units around the Switch.

3. Naming: Click on each PC label and rename them following the convention: S11_01_A, S11_01_B, S11_01_C and S11_01_D.

Step 2: Hardware Identification

1. Click on the Switch and select the Physical tab.
2. Observe the ports. Note that unlike the Hub, a Switch has more ports and may include additional interfaces like console ports, indicating intelligent behavior.

Step 3: Physical Cabling

1. Select the Copper Straight-Through cable (solid black line).
2. Connect S11_01_A (FastEthernet 0) to Switch (Port 0/1).
3. Repeat for the other three PCs using Switch ports 0/2, 0/3 and 0/4.
4. Observation: Notice that the link lights initially turn Orange and then Green. This is because the Switch runs STP (Spanning Tree Protocol), causing a short delay before forwarding begins.

Step 4: Logical Addressing (IP Configuration)

1. Click S11_01_A > Desktop > IP Configuration.
2. Enter Static IP: 192.168.1.1 | Subnet: 255.255.255.0.
3. Configure S11_01_B with IP 192.168.1.2, S11_01_C with IP 192.168.1.3 and S11_01_D with IP 192.168.1.4.

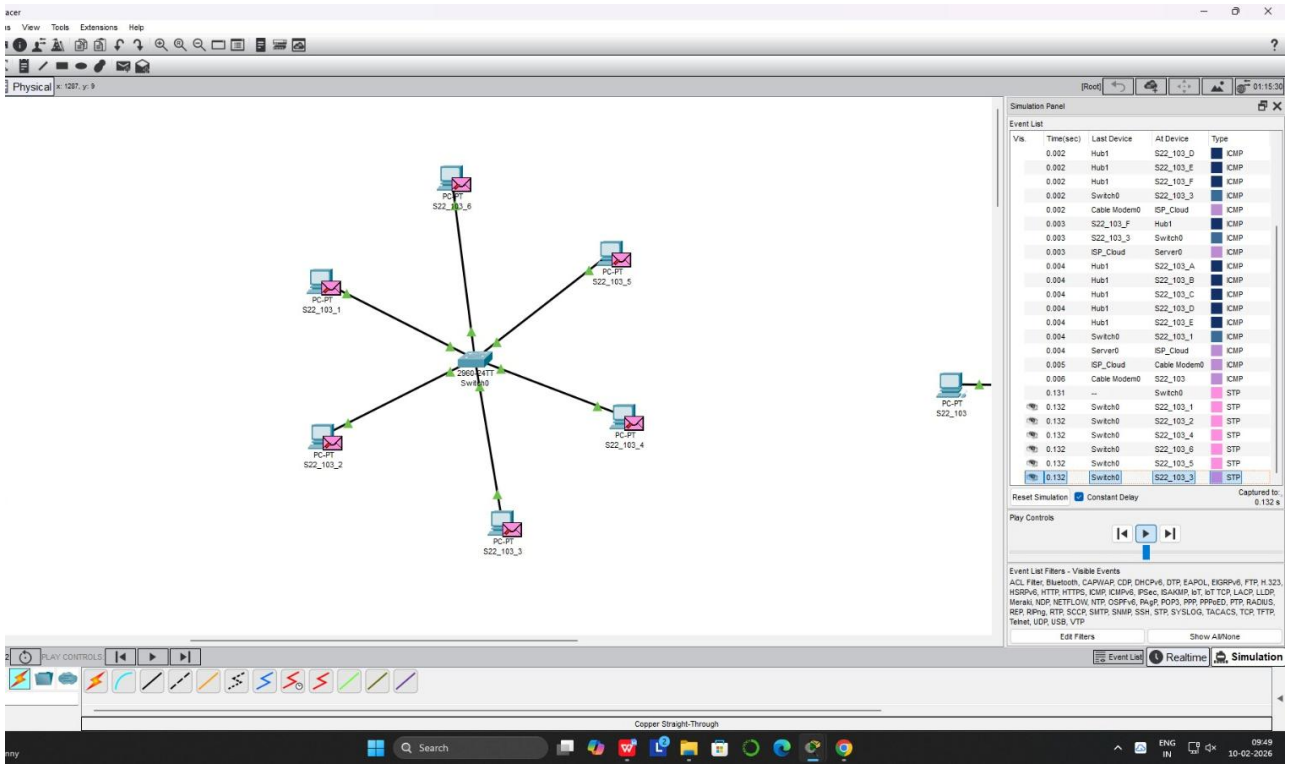
Step 5: Testing and Simulation (Crucial Step)

1. Real-time: Open the Command Prompt on S11_01_A and type ping 192.168.1.3.
2. Simulation Mode:
   o Click the Simulation tab and filter for ICMP. o Use the Add Simple PDU (Envelope) to click S11_01_A then S11_01_C.
   o Click Play and observe carefully: The packet goes to the Switch, and the Switch forwards it only to the intended device after learning MAC addresses.

   o Initially, the Switch may broadcast the packet, but once MAC addresses are learned, it sends data only to PC C.
   o Observe the Green Check on PC C and no unnecessary packet delivery to other PCs after learning.

3. Observations

- Learning Behavior: The Switch learns MAC addresses and builds a MAC table for efficient communication.
- Reduced Traffic: Unlike a Hub, the Switch forwards data only to the intended device, reducing network congestion.
- Initial Broadcast: The first packet may be broadcast, but subsequent communication is direct (unicast).
- STP Delay: Ports take a short time to transition from blocking (orange) to forwarding (green).



Conclusion

The experiment successfully demonstrates the intelligent behavior of a Switch. It concludes that a Switch is a Layer 2 device that uses MAC address-based forwarding, making it more efficient, secure, and reliable than a Hub for modern Local Area Networks.